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Computer Networking HW1

1. Suppose users share a 1 Mbps link. Also suppose each user requires 100 kbps when transmitting, but each user transmits only 10 percent of the time. (See the discussion of statistical multiplexing in Section 1.3.)

   a. When circuit switching is used, how many users can be supported?

   b. For the remainder of this problem, suppose packet switching is used. Find the probability that a given user is transmitting.

   c. Suppose there are 40 users. Find the probability that at any given time, exactly n users are transmitting simultaneously. (Hint: Use the binomial distribution.)

   d. Find the probability that there are 11 or more users transmitting simultaneously.

a. 1 Mbps = 1000 kbps $\frac{1000}{100} = 10$, 10 users.

b. 10%

c. $P(X=n) = \binom{40}{n} \times (0.1)^n \times (0.9)^{40-n}$

d. $P(X \geq 11) = 1 - P(X \leq 10) = 1 - \sum_{n=0}^{10} \binom{40}{n} (0.1)^n (0.9)^{40-n}$

$\approx 1 - 0.9985 = 0.0015$ (0.15%)

2. What are the differences between circuit switching and packet switching? What are the advantages and disadvantages for packet switching, compared with circuit switching?

	Circuit Switching	Packet Switching
資源分配	資源獨佔、頻寬固定	資源共享、動態分配
連線建立	需事前建立連線(call setup)	不需事先建立連線
傳輸效率	即使沒有通訊，依然會佔用頻寬	利用率高，適合突發性流量。
穩定性	穩定、延遲固定	可能發生擁塞延遲or丟包

3. Observe what is a possible packet routing path from a country outside Asia (e.g. US or any EU country) to our school. You can follow following steps to do the observation.

Step 1. Find a free visual traceroute web site from internet.

Step 2. Usually internet free visual traceroute websites would provide an option to let you enter the destination (IP or URL) you would like to view, in which put our school website (www.ntou.edu.tw) to view the traceroute result.

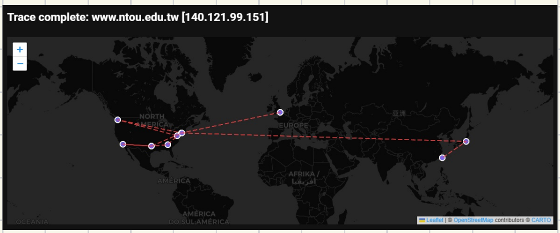
a. what is the visual website you used?

b. what is the path you got?

c. check the global submarine cable map from internet (e.g. https://www.submarinecablemap.com/). Referring to the submarine cable map for those cable connecting to Taiwan and the path you got above, Guess which submarine cables are utilized for the traceroute results according to the path you observed, Why?

a. https://traceroute.to/

b、



Hop	IP	Host	AS	Min ms	Max ms	Average ms	Count
1	10.3.21.14	gw1.hosting.stackcp.net	Private/Reserved	2.85	11.78	5.11	4
2	45.8.227.251		48254 - VPS Platform	2.98	3.41	3.17	4
3	185.146.164.65	et1-1-5-cr1-lon-thn.as48254.net	48254 - UK THN INFRA	3.69	3.88	3.79	4
4	212.119.4.228	xe-0-4-2-2.a03.londen12.uk.bb.gin.ntt.net	2914 - Lon Backbone	4.51	5.47	4.81	4
5	129.250.5.244	ae-11.r22.londen12.uk.bb.gin.ntt.net	2914 - NTT America, Inc.	4.61	15.64	7.54	4
6	129.250.6.147	ae-7.r22.nwrknj03.us.bb.gin.ntt.net	2914 - NTT America, Inc.	204.89	205.5	205.06	4
7	129.250.3.49	ae-11.a02.nycmny17.us.bb.gin.ntt.net	2914 - NTT America, Inc.	80.6	102.05	86.08	4
8	129.250.8.230	ae-0.arelion.nycmny17.us.bb.gin.ntt.net	2914 - NTT America, Inc.	81.31	125.3	97.71	4
9	Request timed out						4
10	62.115.136.201	ash-bb2-link.ip.twelve99.net	1299 - Arelion Sweden Ab	79.97	80.24	80.11	4
11	62.115.137.133	atl-bb2-link.ip.twelve99.net	1299 - Arelion Sweden Ab	135.95	153.41	140.47	4
12	62.115.143.237	atl-b24-link.ip.twelve99.net	1299 - Arelion Sweden Ab	135.83	136.25	90.69	4
13	62.115.116.212	dls-b7-link.ip.twelve99.net	1299 - Arelion Sweden Ab	135.7	136.57	136.24	4
14	62.115.140.246	lax-bb2-link.ip.twelve99.net	1299 - Arelion Sweden Ab	136.18	151.06	139.98	4
15	Request timed out						4
16	62.115.14.42	hurricane-ic-371590.ip.twelve99-cust.net	1299 - Arelion Sweden Ab	137.33	139.36	138.20	4
17	184.104.189.162	be48.core1.lax2.he.net	6939 - Hurricane Electric	258.36	258.62	258.51	4
18	Request timed out						4
19	Request timed out						4
20	184.105.80.142	100ge0-80.core2.tpe1.he.net	6939 - Hurricane Electric	255.9	256.99	192.37	4
21	65.49.108.2	chief-telecom-inc.port-channel4.switch2.tpe1.he.net	6939 - Hurricane Electric LLC	255.85	256.24	256.05	4
22	192.192.68.63		1659 - Imported Inetnum Object for Moec	256.14	256.54	256.37	4
23	192.192.61.53		1659 - Imported Inetnum Object for Moec	256.97	257.14	257.02	4
24	192.192.61.89		1659 - Imported Inetnum Object for Moec	257.64	259.66	258.24	4
25	Request timed out						4
26	140.121.249.245		1659 -	258.31	258.52	258.40	4
27	Request timed out						4
28	Request timed out						4
29	Request timed out						4
30	Request timed out						4

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c、

1、 2、

從 UK倫敦出發 → US紐約 → US西雅圖 → US華盛頓 → US亞特蘭大 → US洛杉磯 → US達拉斯 → US紐約 → JP東京 → TW臺北

3、 4、

1、 ISP：NTT
Delay：≈205.5ms
Route：倫敦 ⟷ 美國東岸
Guess： Amitie

2、 ISP：Arelion / HE
Delay：≈ 80 ~ 140ms
Route：美國境內
→ 美國境內骨幹網路，非海底電纜。

3、 ISP：Arelion / HE
Delay：255ms
Route：洛杉磯 ⟷ 紐約 ⟷ 東京
→ 由於紐約 ⟷ 東京沒有直連海纜，且由美東出海進亞洲非常奇怪。在缺乏更多資訊下猜測：
AEC-1 → AAE-1 → SJC2 回東京。

4、 ISP：MOEC (TANet)
Delay：255ms
Route：東京 ⟷ 臺北
Guess：APCN-2

4. What is the internet access technology you use at home (e.g. ADSL, cable modem, …)? a. What is the maximal bandwidth your internet access ISP announced? b. Do a bandwidth testing (you can find many tools/services for internet speed test), and check do your internet connect can reach the maximal bandwidth as your ISP claimed? If not, what is the possible reasons? c.

a. ISP宣稱使用光纖同軸混合(Hybrid Fiber Coaxial/HFC)技術，這屬於cable modem的接入方式。

b. ISP宣稱下載120Mbps/上載40 Mbps

c.

```
whitebear13579@poweredge-r410:~$ speedtest

   Speedtest by Ookla

[error] Error: [0] Timeout occurred in connect.
      Server: Taipeinet - Taipei (id: 7429)
         ISP: Peicity Digital Cable Television.
Idle Latency:     6.53 ms   (jitter: 0.92ms, low: 5.60ms, high: 7.92ms)
    Download:   125.45 Mbps (data used: 89.5 MB)
                138.93 ms   (jitter: 40.73ms, low: 7.91ms, high: 336.09ms)
      Upload:    41.34 Mbps (data used: 20.2 MB)
                 19.83 ms   (jitter: 1.65ms, low: 10.22ms, high: 24.35ms)
 Packet Loss:     0.0%
  Result URL: https://www.speedtest.net/result/c/dd7bea14-7696-4539-8a8d-e7bebcf2ec9f
```

測試結果顯示，下載為125.45 Mbps；上載為41.34 Mbps，均超過ISP宣稱的數值。

測量值高於宣稱值的原因為，ISP會進行over-provisioning，來確保使用者在測速時必能達到標稱速度，或者是確保尖峰期間網路品質的穩定。

5. Please check with some job hunting/matching websites (e.g. 104, or 1111) and find three companies related in network/telecommunication technique. What kinds of network related knowledge are preferred in these companies?

1、中華電信：網路工程師、資安網路工程師……、

2、神準科技：IoT軟韌體工程師、網通產品研發工程師……、

3、台灣大哥大：企業網路工程師、雲端網路工程師……、

網路相關知識：

a、基礎網路技術：TCP/IP、Routing、Switching

b、網通設備：Router、Switch、Access Point

c、網路資安：Firewall、IPS/IDS、WAF

d、大型網路：BGP、MPLS、OSPF